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The Future of Work. Now.™

From promising to proven: Good practice for machine learning in flotation

Lidia Auret



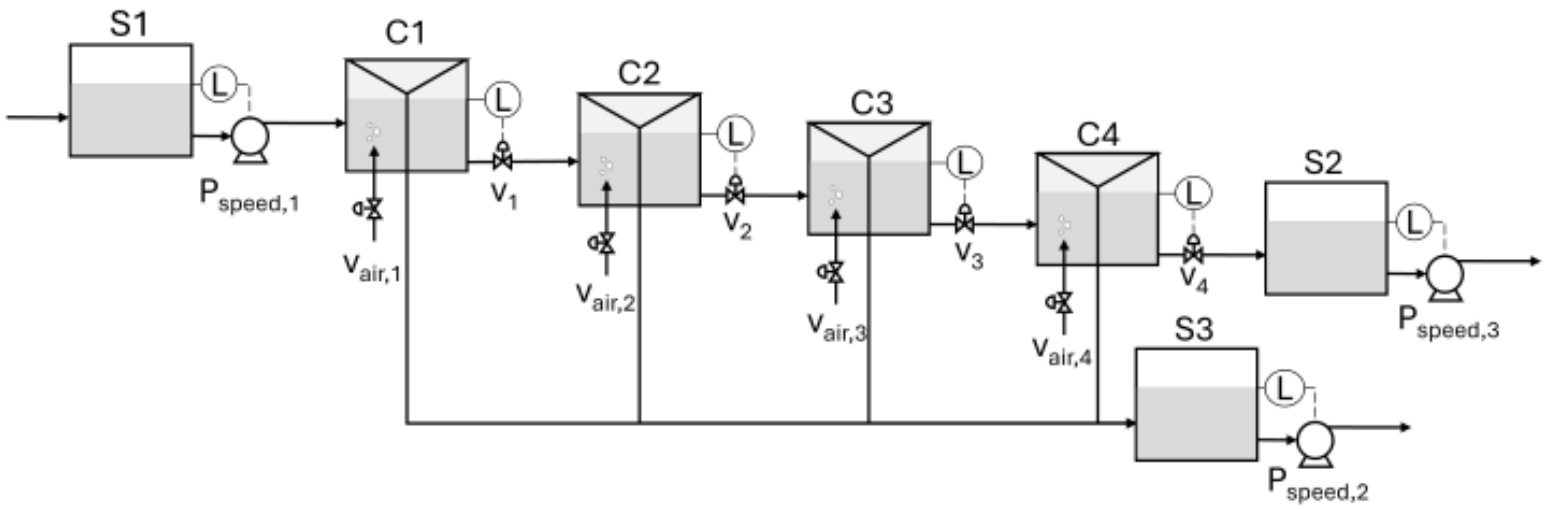
bsi ISO/IEC 27001 Information Security Management CERTIFIED ISO/IEC 27701 Privacy Information Management CERTIFIED

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Flotation ML: A case study

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- Copper rougher bank
- 21 days of operational data at 1 min sampling (30 240 samples)
- Varying disturbances, closed-loop level control
- Operator interventions



Group	Variable
Operator inputs (measured)	Air valves 1 and 2 (%)
	Air valves 3 and 4 (%)
	Frother pump speed (%)
Disturbances (measured)	Feed rate (t/h)
	Feed grade (% Cu)
Disturbances (unmeasured)	Share of fast-floating Cu mineral (%)
Intermediate (measured)	Mean pulp bubble size (mm)
	Mean superficial gas velocity (cm/s)
	Mean tail valve (%)
	Mean cell pulp level (m)
	Conc sump level (m)
	Tail sump level (m)
Outputs	OSA conc grade (% Cu)
	OSA tail grade (% Cu)
	Two-product formula recovery (%)

https://github.com/LidiaAuret/flotation-ml-good-practice

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Flotation ML: A case study

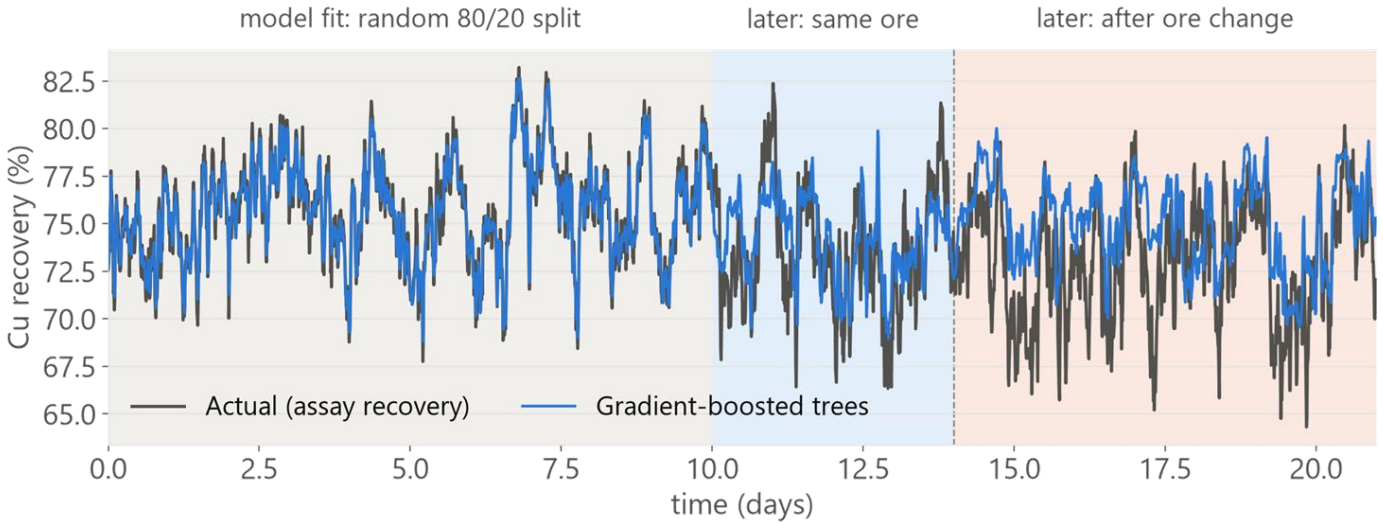
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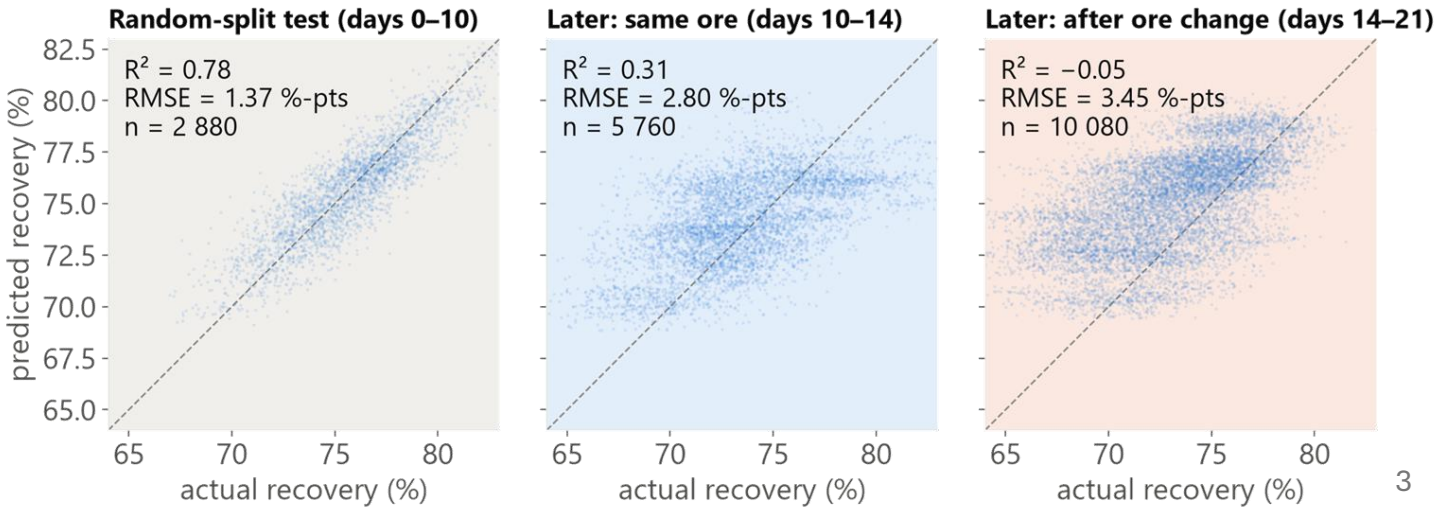
- ML model: Gradient-boosted trees

Random training/validation/test split
“Good” performance

Model performance **mediocre** for future period of similar operation

Model performance **very poor** for future period of changed conditions





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Flotation ML: A case study

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- What went wrong? Data leakage and lack of reproducibility!

Kapoor and Narayanan (2023) found **329 papers** with machine learning application errors in 17 research fields, leading to widely optimistic conclusions

Lack of clean separation between training and test sets	Illegitimate features	Test set not representative
• No test set • Pre-processing on training and test sets • Feature selection on training and test sets • Duplicates	• E.g., proxy for output variable included in inputs • Domain-specific	• Temporal leakage (shuffling) • Nonindependence between train and test sets • Sampling bias

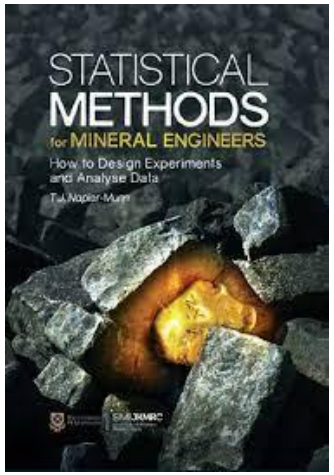
Kapoor, S., Narayanan, A., 2023. Leakage and the reproducibility crisis in machine-learning-based science. Patterns 4, 100804.

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Flotation ML: Best practices

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- Illegitimate features: Proxy for output included in input variables
  - Ratio problem (Napier-Munn), e.g., two-product formula calculation
  - Where feed grade variable was used to calculate recovery



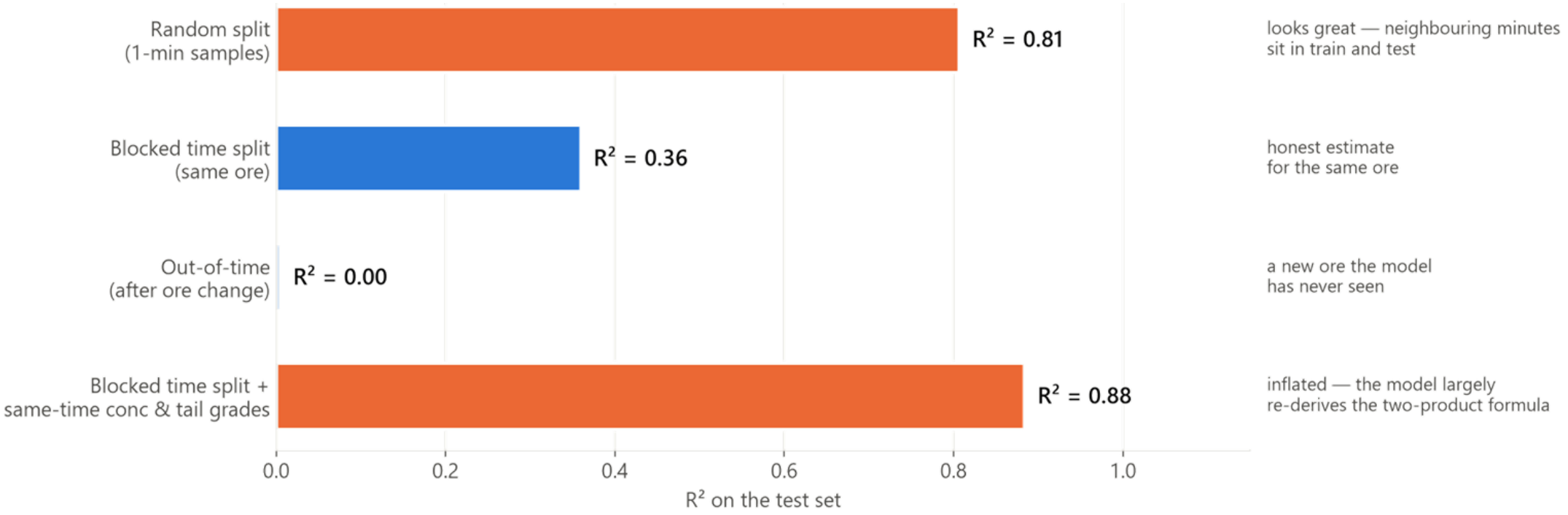
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Flotation ML: Best practices

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- Test set not representative: Temporal leakage

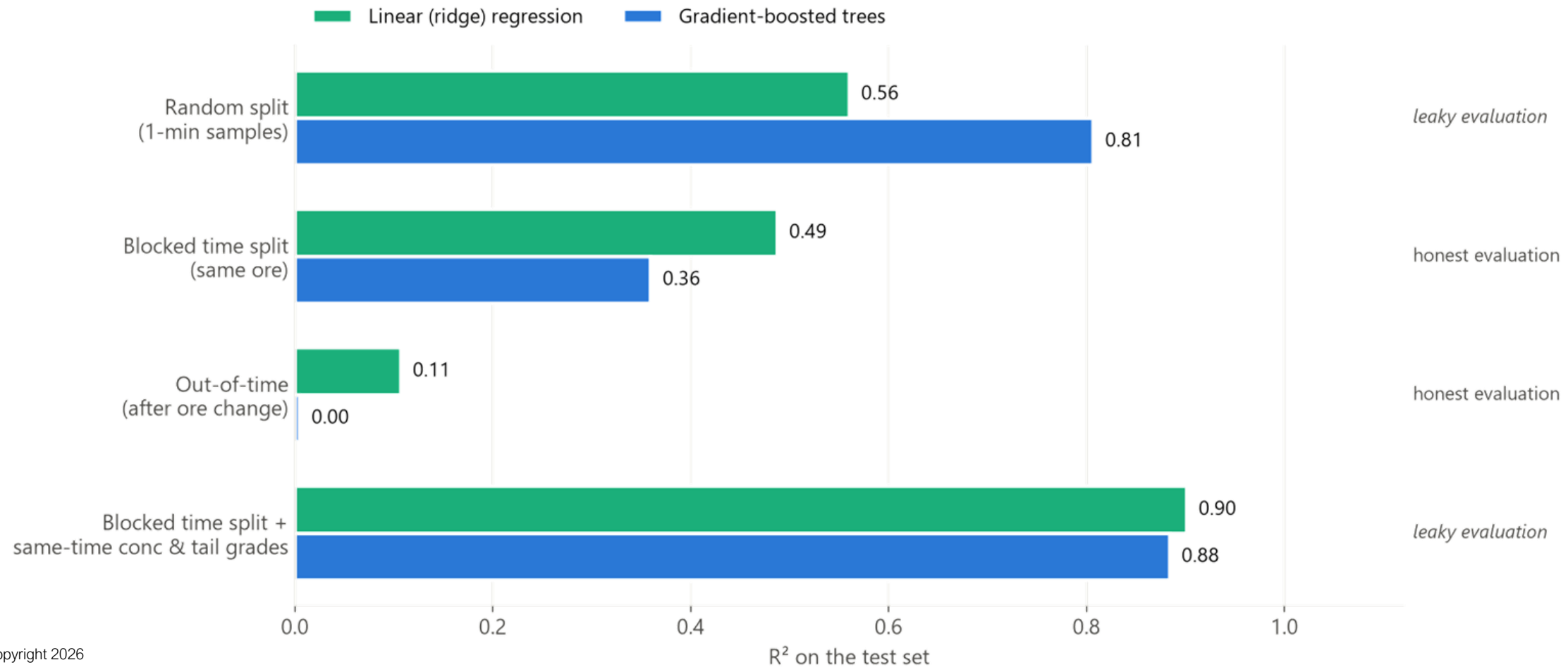


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Flotation ML: Best practices

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- Test set not representative: Temporal leakage
  - Complex models more susceptible…

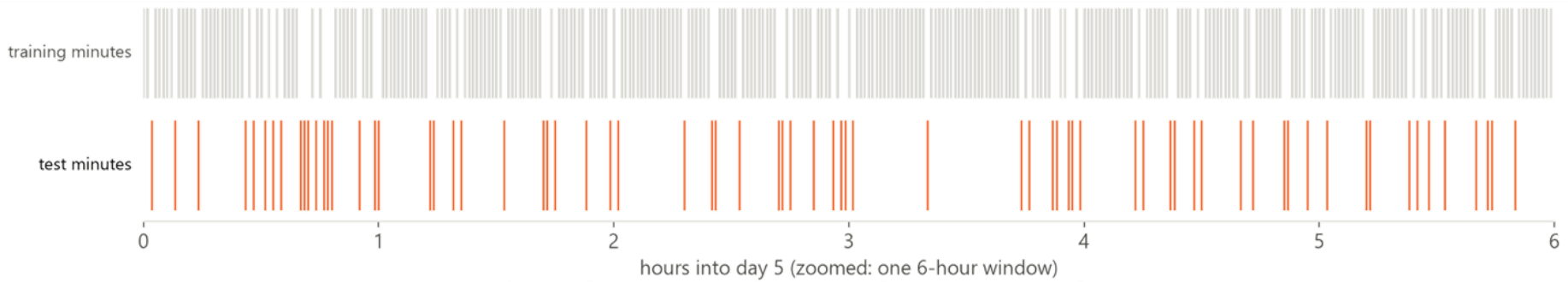


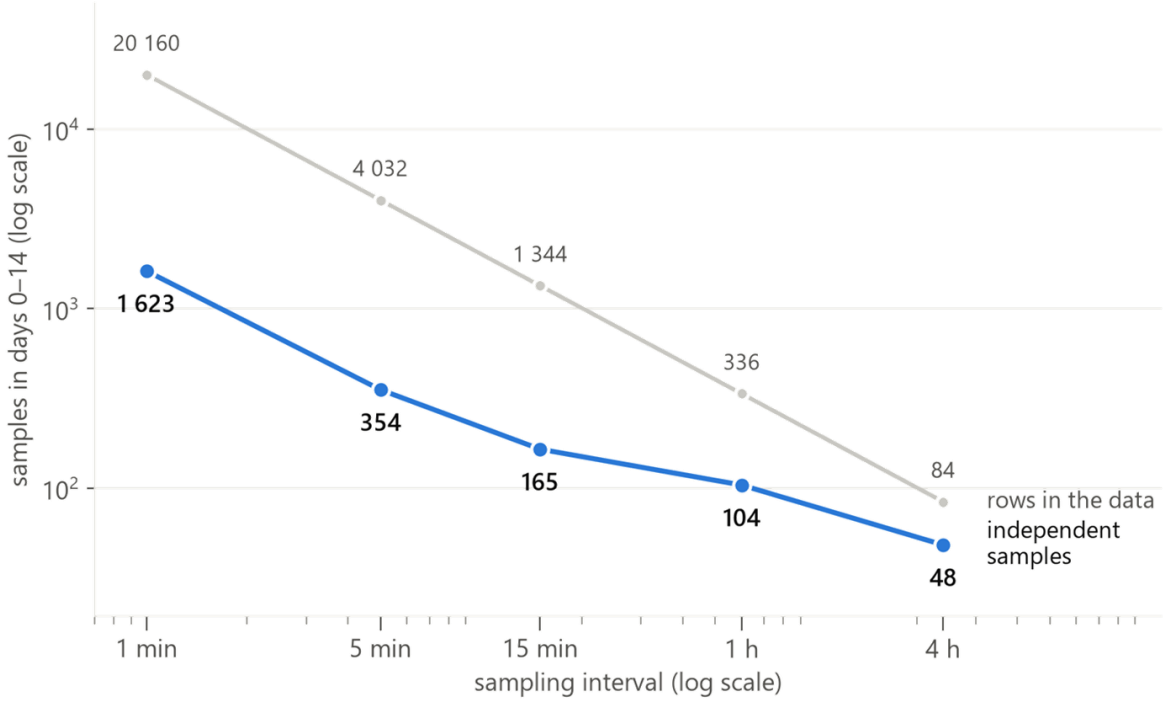
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Flotation ML: Best practices

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- Test set not representative: Temporal leakage



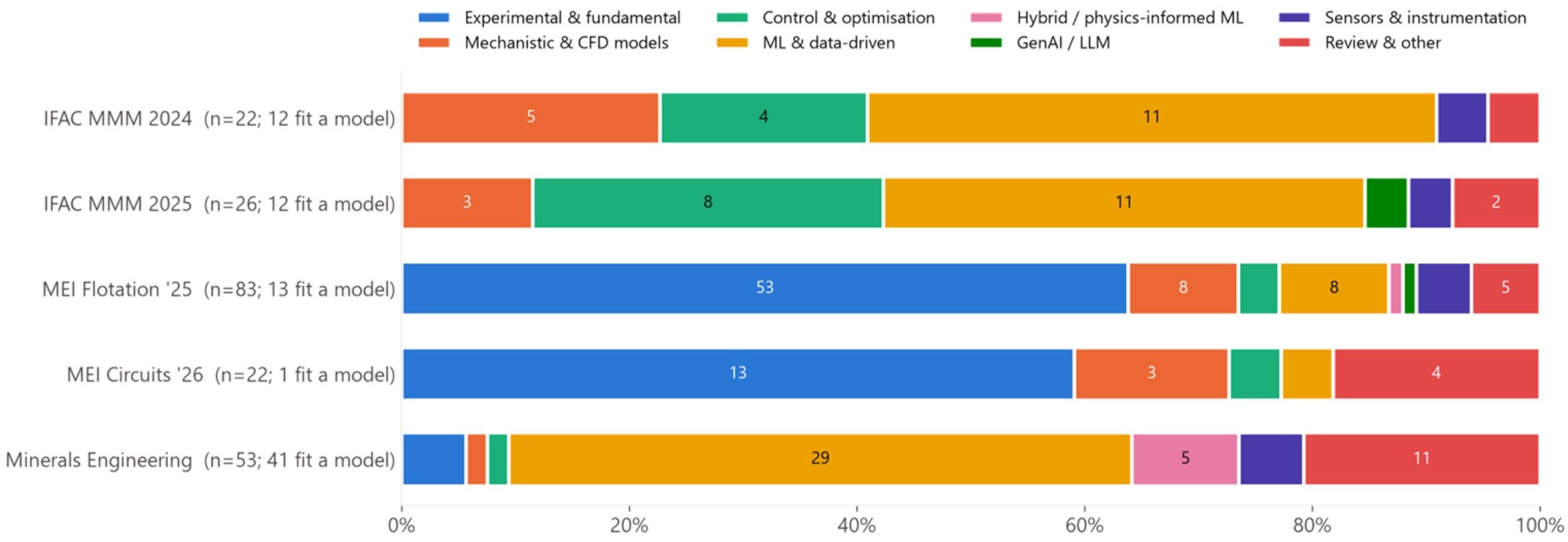


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Flotation ML: A survey

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- How do we fare in flotation ML…?



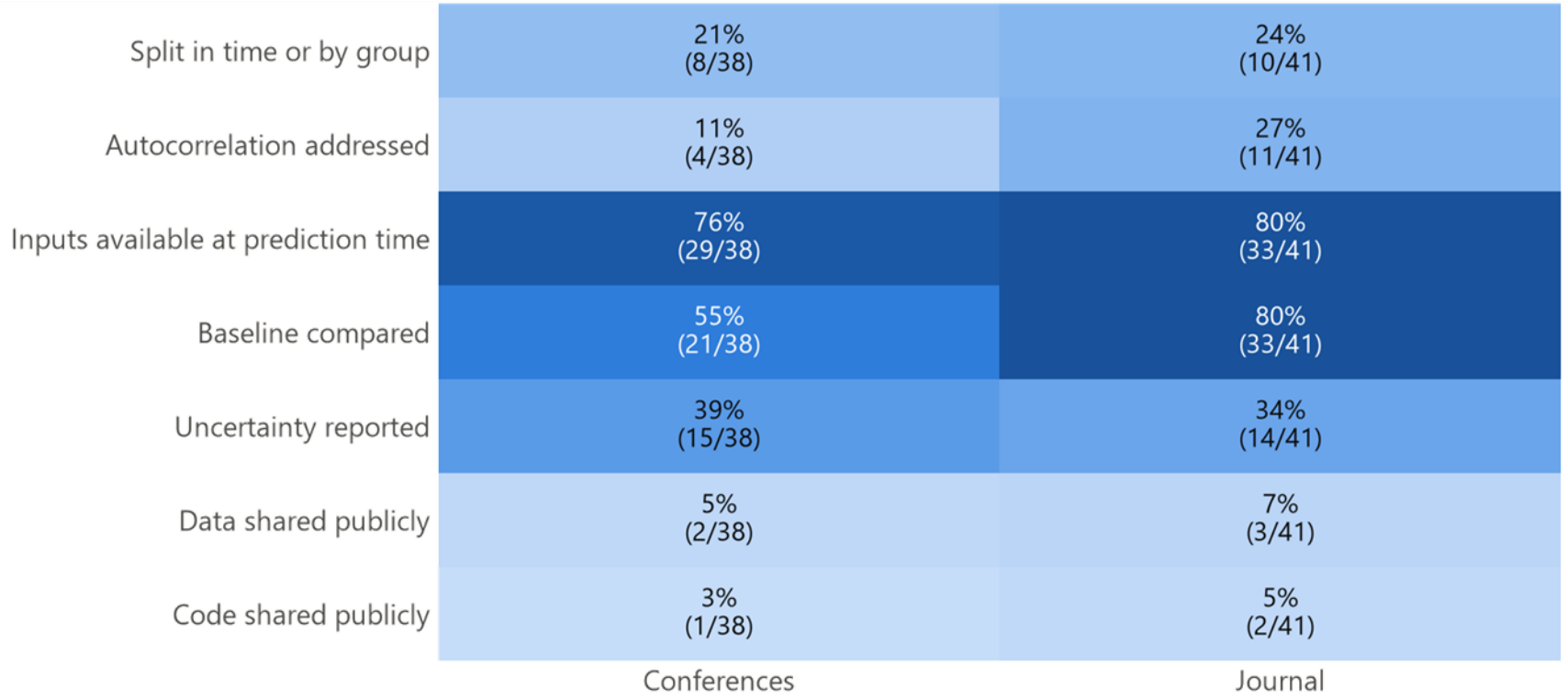
Source: full-text assessment of conference and journal papers, coded with AI assistance; 49 of 206 papers verified by the author. Authors anonymised.
79 papers fit a data-driven model; 13 of them have a non-ML primary method. MEI selections = flotation & circuits papers; journal = 'flotation + machine learning' search.

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Flotation ML: A survey

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- How do we fare in flotation ML in terms of data leakage/reproducibility…? Room for improvement!



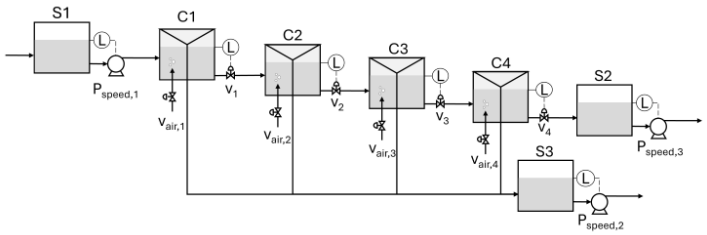
Source: full-text assessment of conference and journal papers, coded with AI assistance; 42 of 79 data-driven papers verified by the author. Authors anonymised.

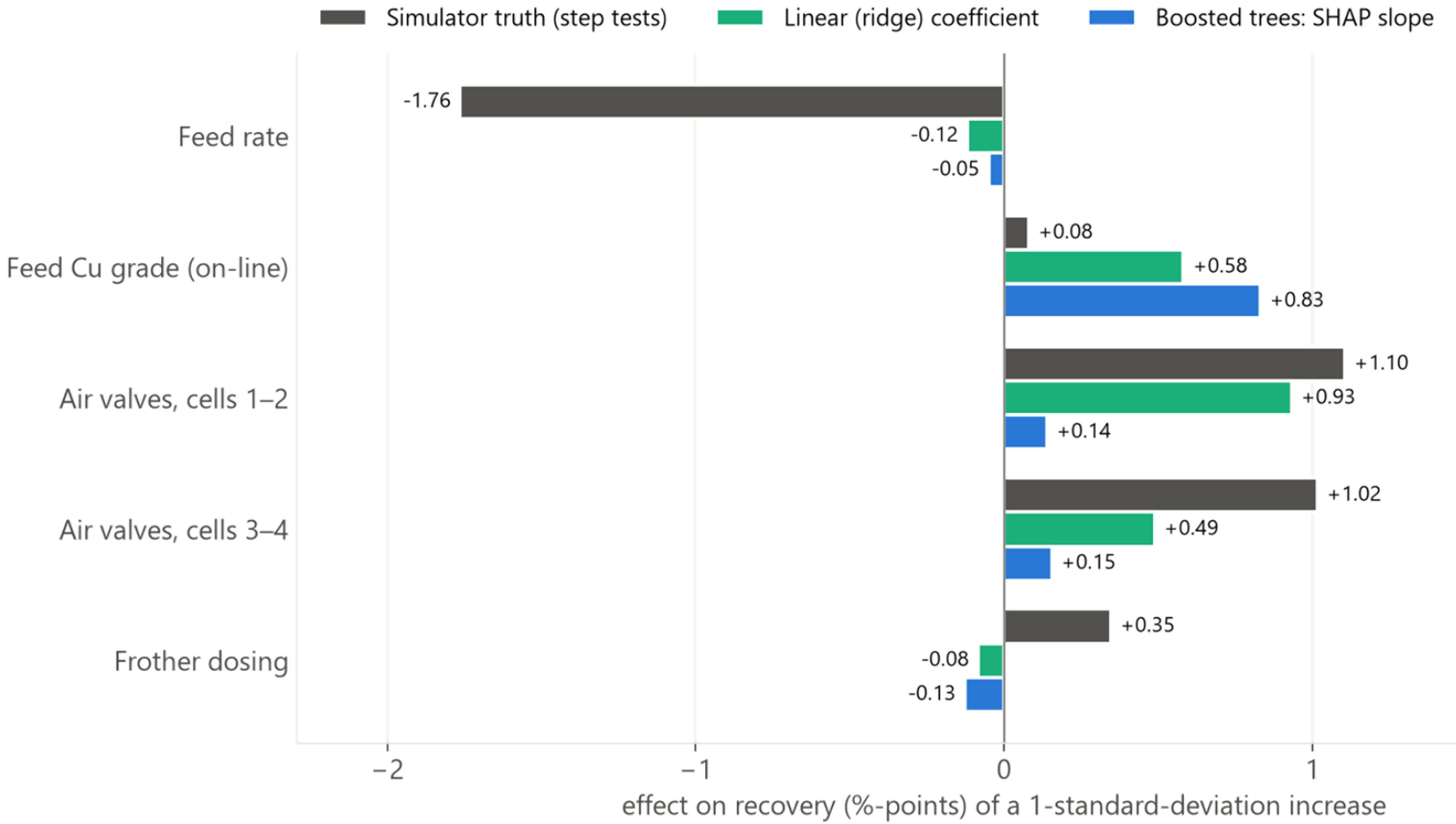
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Flotation ML: A case study

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- Explainable AI techniques, e.g., SHAP
  - Explains the model, not the phenomenon



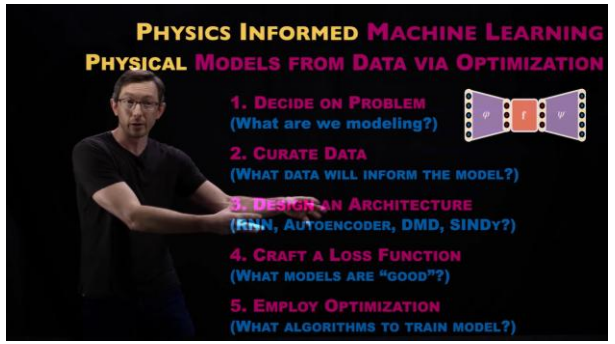


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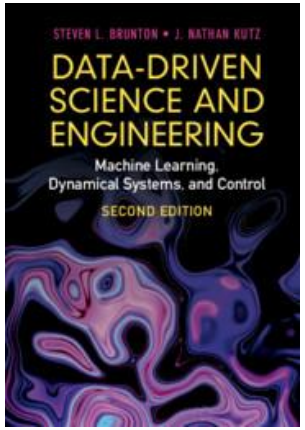
Flotation ML: A case study

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- What went wrong? Lack of physics in modelling approach.
- Embed physics (fundamental/phenomenological models) in ML:
  - 1. Decide on problem
  - 2. Curate data
  - 3. Design an architecture
  - 4. Craft a loss function
  - 5. Employ optimization



Course: Physics Informed Machine Learning by Steve Brunton (YouTube)



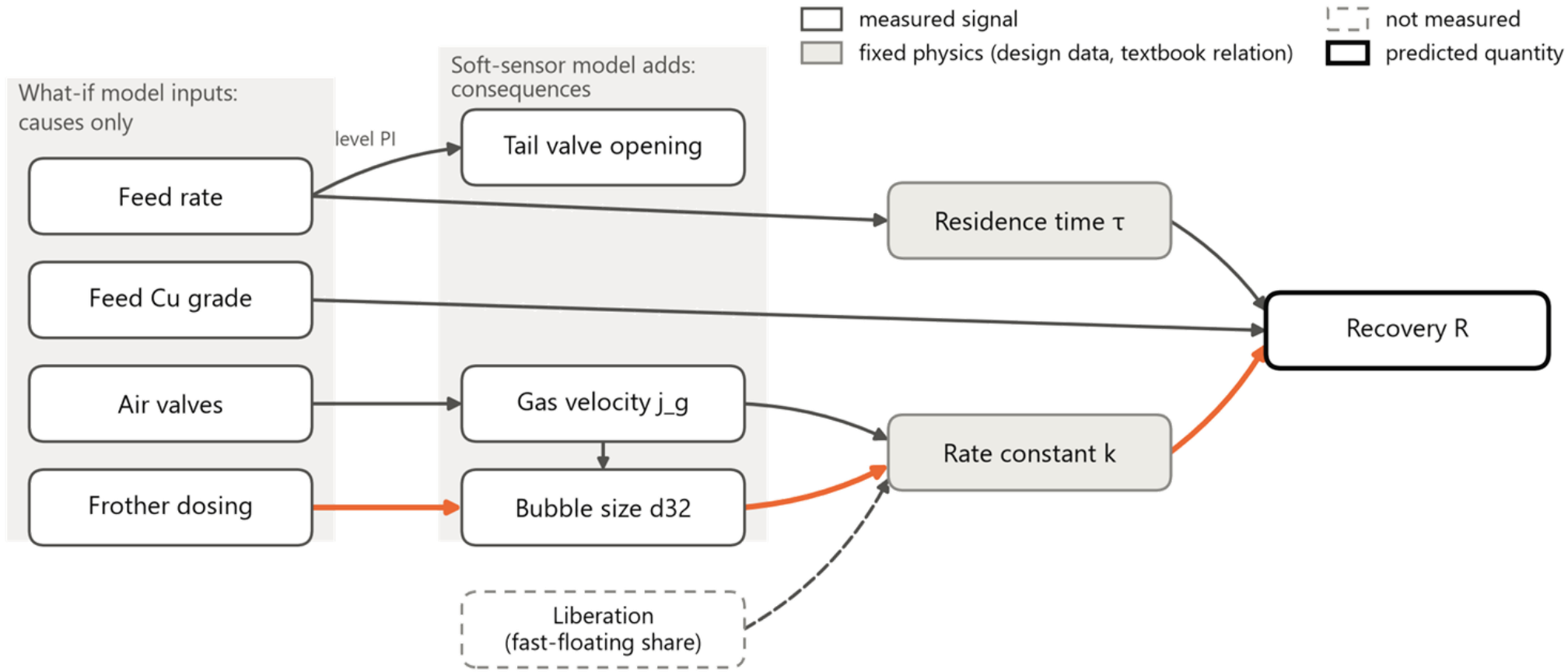
Brunton, S.L., Kutz, J.N., 2022. Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control. Cambridge University Press.

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Flotation ML: Best practices

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- 1. Decide on the problem
  - E.g., causal model should not include intermediate variables

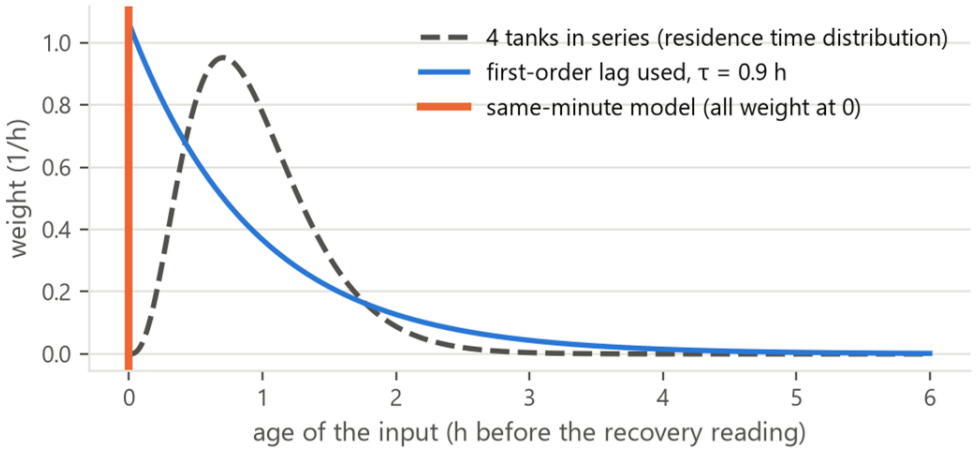


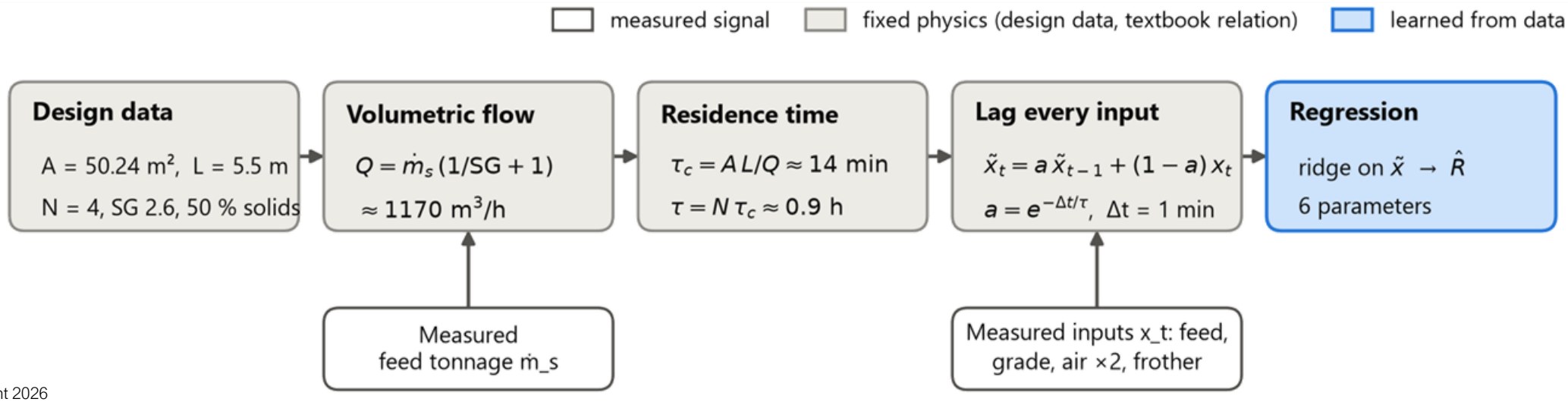
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- 2. Curate data
  - E.g., residence time



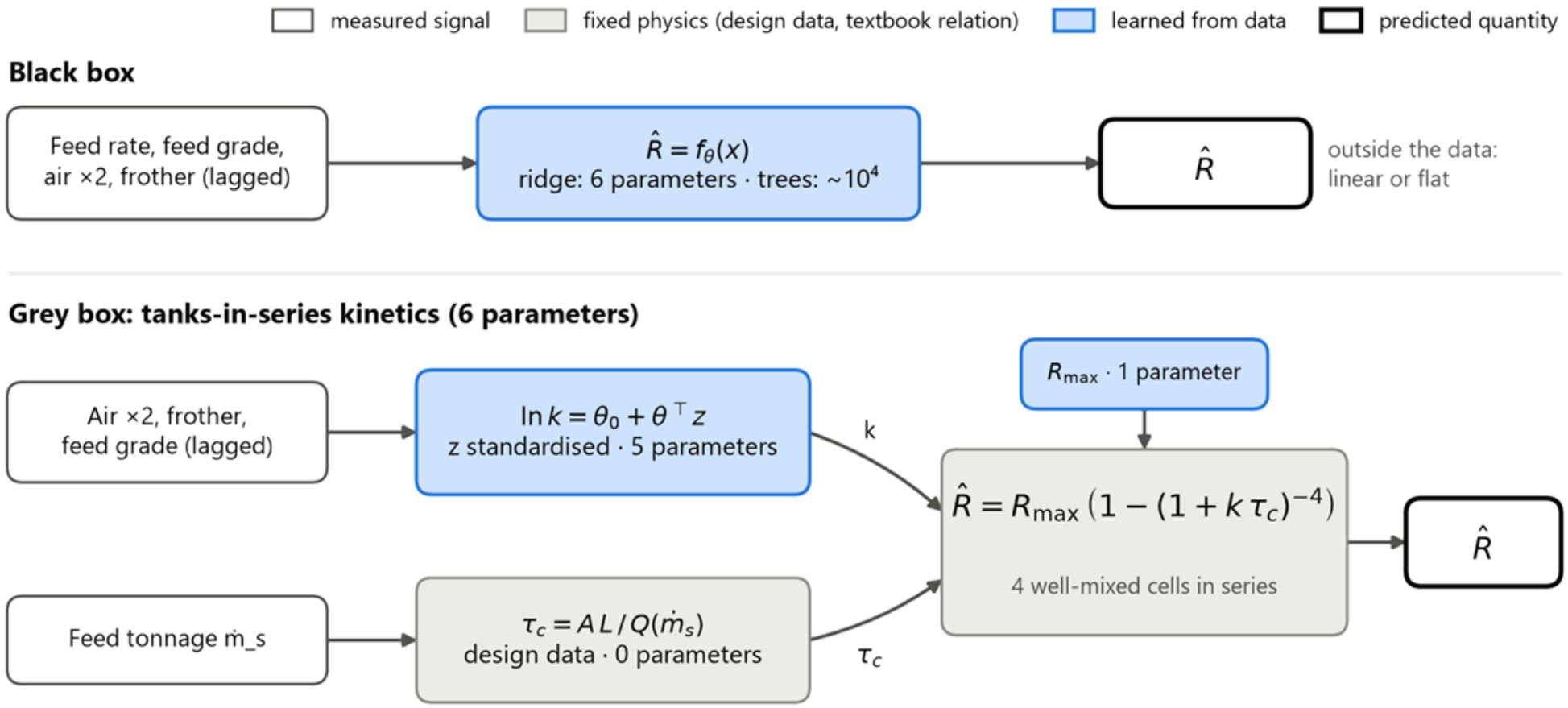


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- 3. Design an architecture

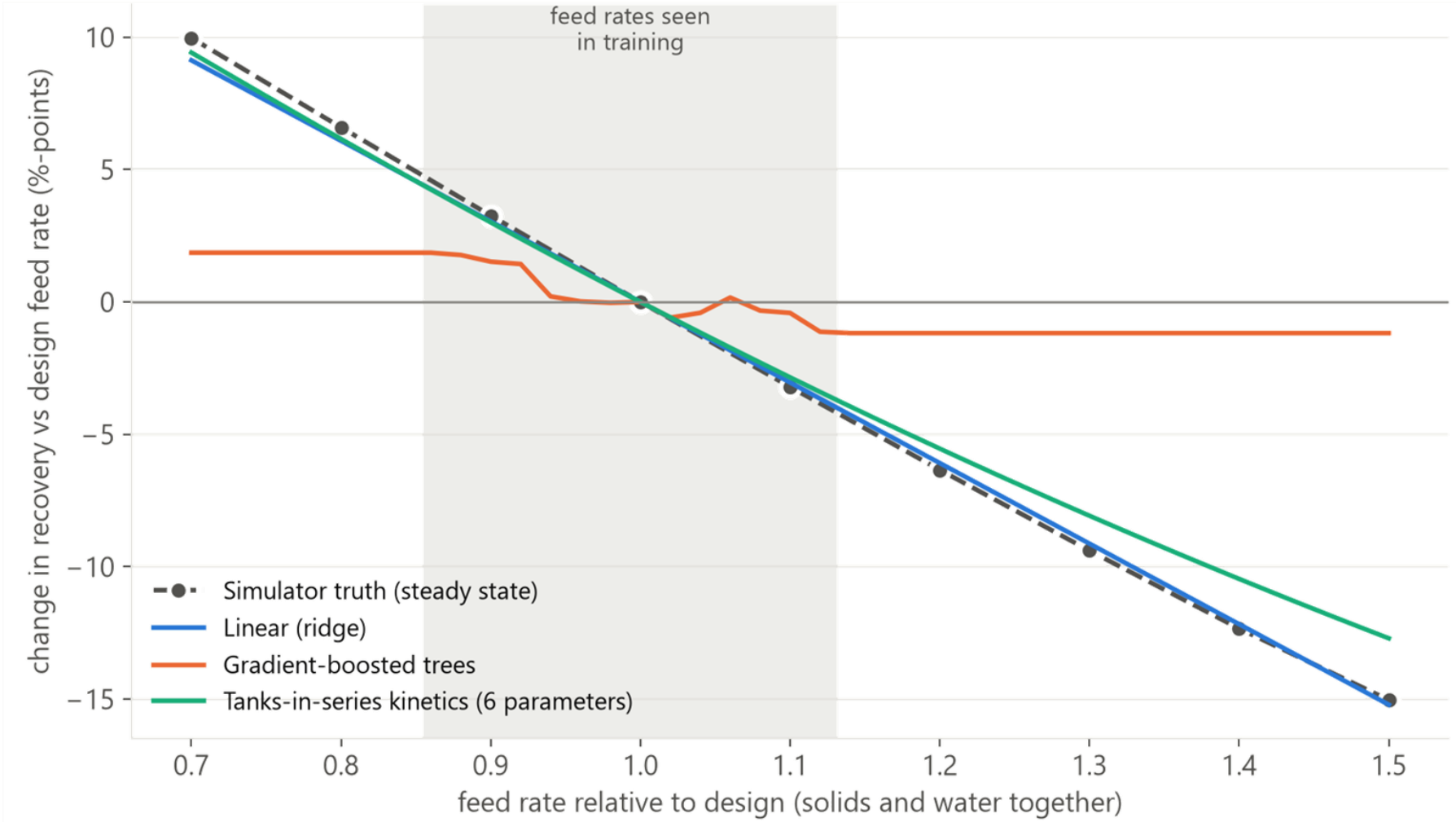


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Flotation ML: Best practices

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- 3. Design an architecture



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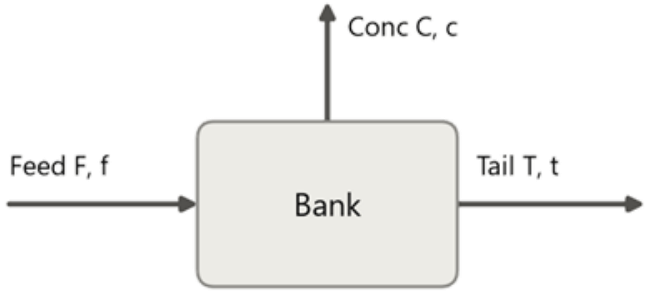
Flotation ML: Best practices

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- 4. Craft a loss function

measured signal | fixed physics (design data, textbook relation) | learned from data

Two-product mass balance



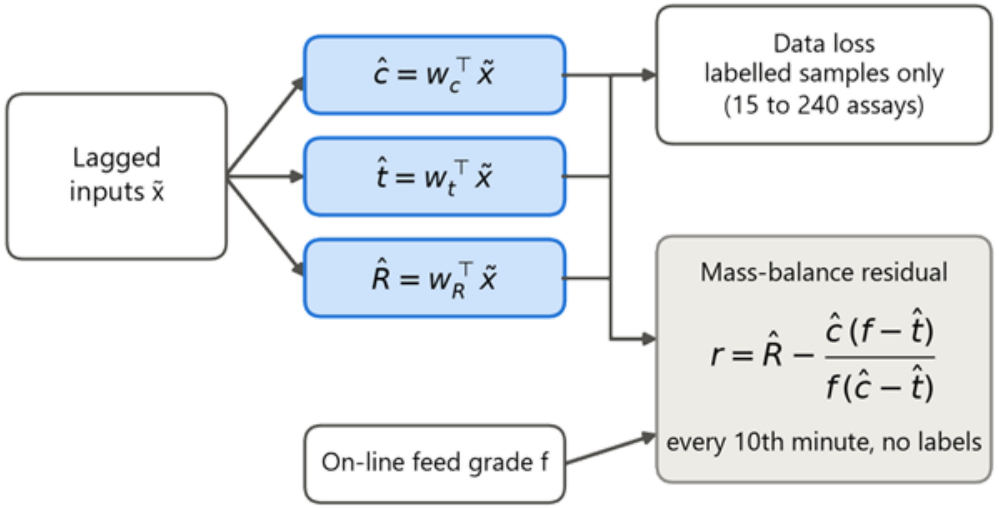
Solids and Cu balances

$$F = C + T$$

$$Ff = Cc + Tt$$

$$\Rightarrow R = \frac{Cc}{Ff} = \frac{c(f-t)}{f(c-t)}$$

Training with the balance as a penalty



Loss minimised (L-BFGS)

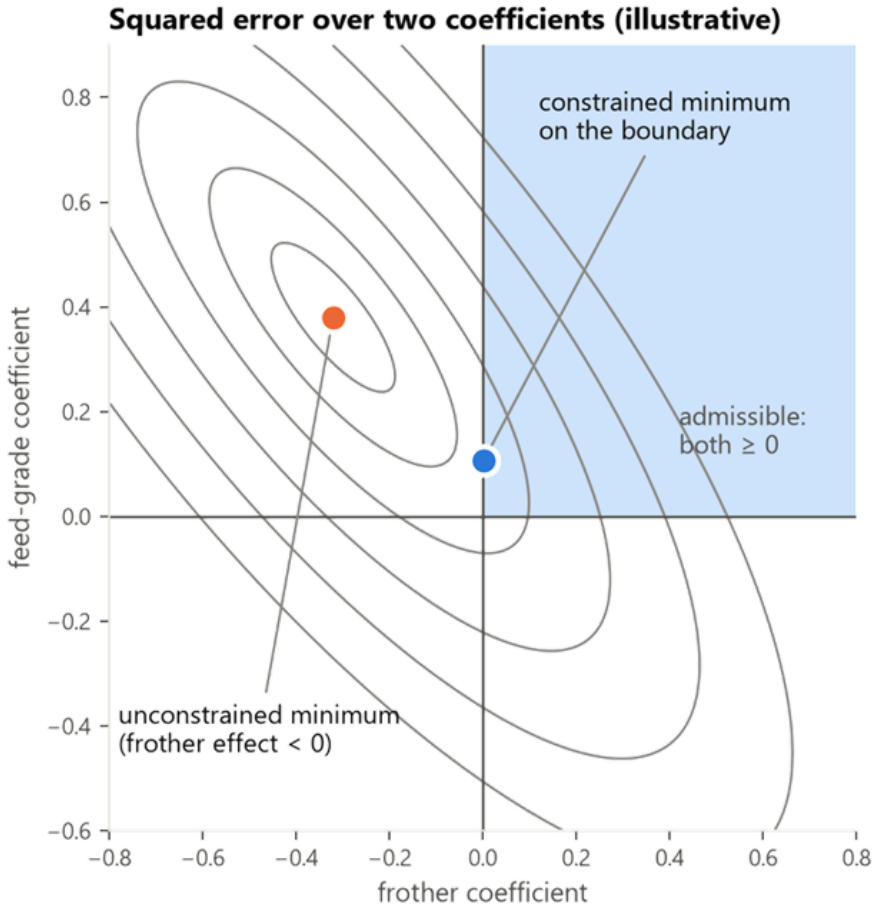
$$\mathcal{L} = \sum_{o \in \{c,\, t,\, R\}} \frac{1}{n_\ell} \sum_{\ell} \left(\frac{\hat{y}_o - y_o}{\sigma_o} \right)^2 + \lambda \frac{1}{n_u} \sum_{u} \left(\frac{r}{\sigma_R} \right)^2$$

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Flotation ML: Best practices

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- 5. Employ optimization



fixed physics (design data, textbook relation) learned from data

Bounded-variable least squares (scipy lsq_linear)

$$\min_{\beta} \ \|y - \beta_0 - \tilde{X}\beta\|^2 + \lambda \|S\beta\|^2$$

subject to $\beta_{\text{feed rate}} \leq 0$

$\beta_{\text{air 1-2}}, \ \beta_{\text{air 3-4}}, \ \beta_{\text{frother}}, \ \beta_{\text{feed grade}} \geq 0$

S: plant-wide input standard deviations, so the penalty is per typical move

Stage 4 · soft constraint

violations cost extra loss

but remain possible

weight λ must be chosen

Stage 5 · hard constraint

inadmissible coefficients

are never visited

no weight to choose

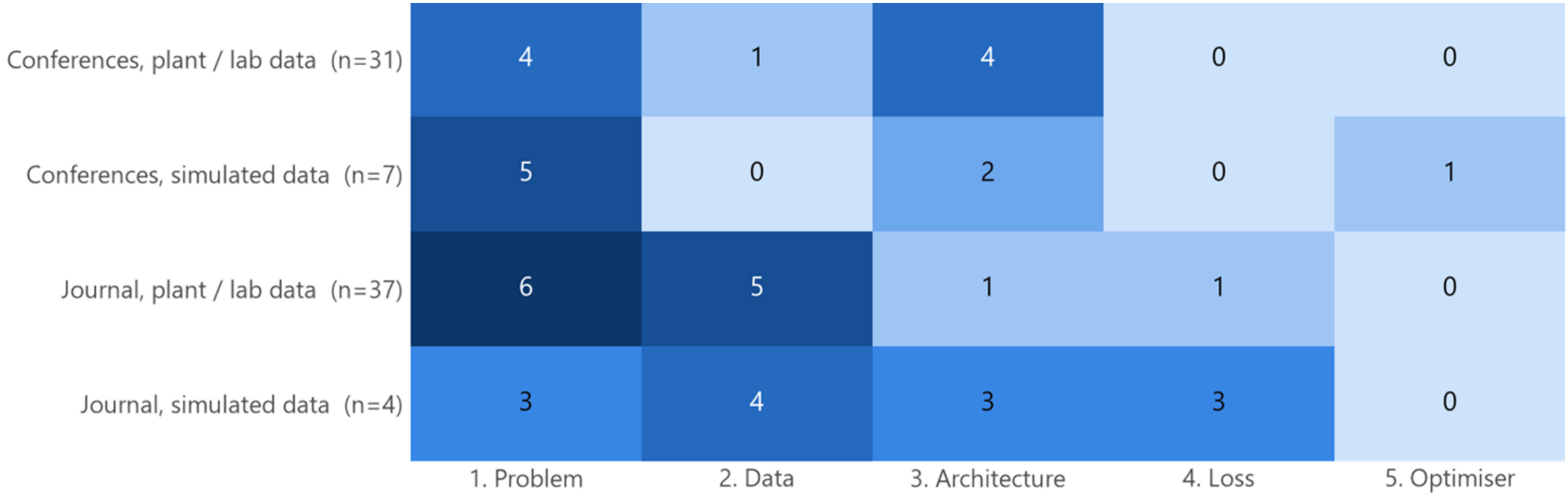
Result on one-day windows: wrong-sign coefficients 12 → 0, next-day error 2.43 → 2.44 %-points.

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Flotation ML: A survey

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- How do we fare in flotation ML in terms of physics informed ML…? Room for improvement!



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Flotation ML: Best practices

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- “Is it worth it?”
  - Define upfront what good enough model performance is
  - Always compare to a simple ML baseline
  - Preferably compare to a fundamental/phenomenological/semi-empirical baseline
  - Choose the simplest, fastest, most interpretable model that meets the required performance criteria

- Report uncertainty and variability
  - Most ML models are non-deterministic: do repeated fits and report variability

- Report hyperparameter optimization
  - And spend the same effort on all contenders

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Conclusions

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- The problem of data leakage, lack of reproducibility, and lack of physics informed ML:
  - Prevalent, including in flotation ML
  - Erodes trust and hinders innovation transfer to industry
- Let's work together to apply ML techniques with the same rigour as other scientific methods
  - In your own research
  - Reviewing others' research
  - In your prompt engineering!
- Toolbox:
  - Kapoor and Naranayan (2023) data leakage checklist and model sheet
  - Brunton and Kutz (2022) physics informed ML stages for inspiration
  - Sufficient detail for verification
  - Shared data and simulations

https://github.com/LidiaAuret/flotation-ml-good-practice

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An invitation

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https://conferences.ifac-control.org/cca-mmm-2027/

MMM CCA 2027

SACAC SOUTH AFRICAN COUNCIL FOR AUTOMATION AND CONTROL

IFAC INTERNATIONAL FEDERATION OF AUTOMATIC CONTROL

4th CCA 2027 • 8th MMM Workshop

6–8 September 2027 • Tshikwalo Game Lodge, Dinokeng, Pretoria, South Africa

IMPORTANT DATES

Submissions open:	**9 November 2026**
Submission deadline:	**26 March 2027**
Notification of acceptance:	**25 May 2027**
Final paper due:	**27 June 2027**
Early registration:	**5 July 2027**

TOPICS

CCA

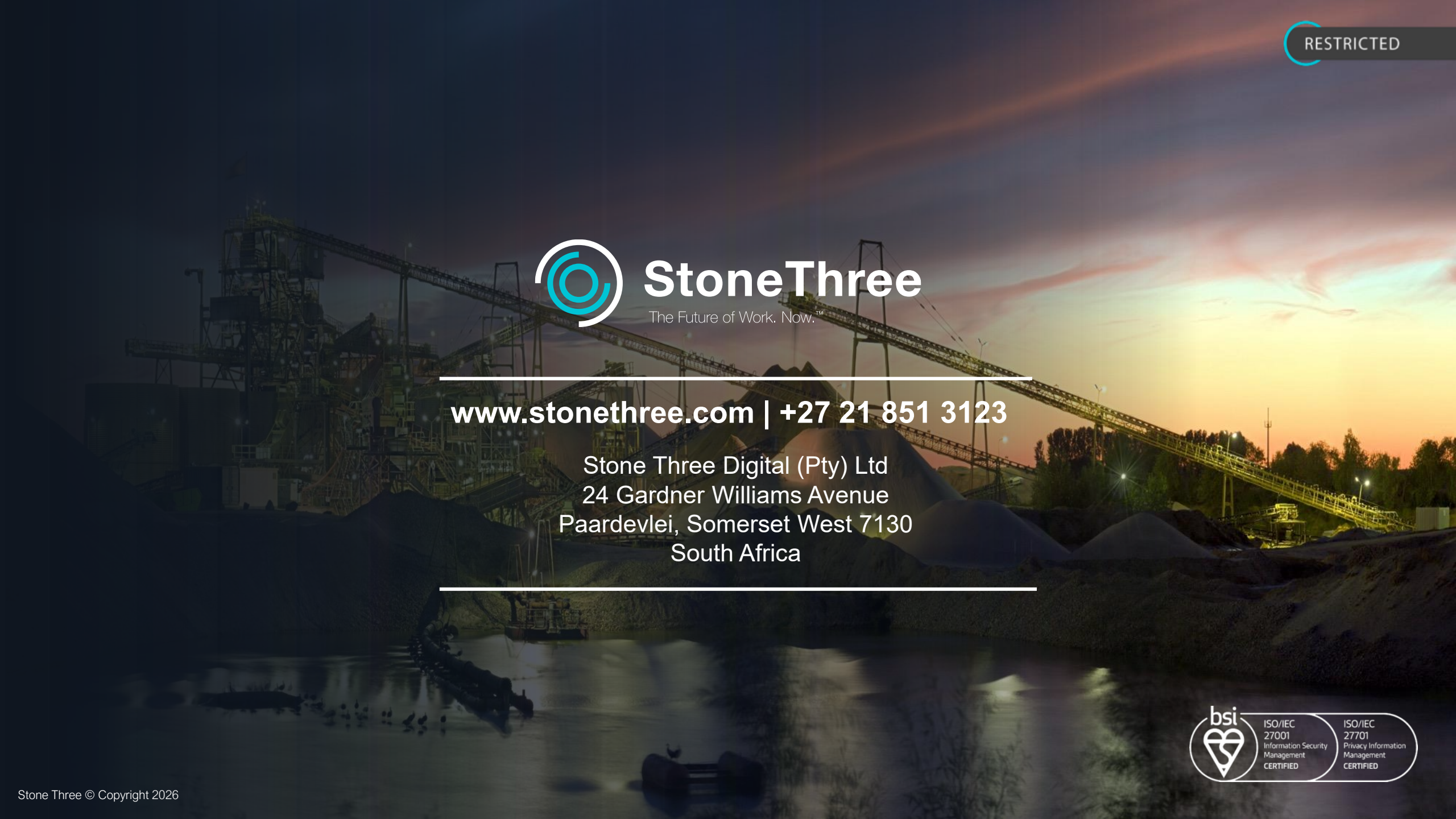
- Automotive & aerospace systems
- Modelling & system identification
- Optimisation & automation
- Robotic systems
- Process control
- Novel control theory
- Education in process control & automation

MMM

- Automation & control in mining
- Mineral processing & beneficiation
- Metallurgical processes & smelting
- Sensor technology & instrumentation
- Safety and environmental concerns
- Case studies & industrial applications
- AI in mining, minerals and metals

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